EXP 30: Prolog Program for Backward Chaining

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Aim:

To develop a Prolog program that demonstrates backward chaining, so that a goal is proved by recursively reducing it to subgoals until they match known facts.

Algorithm:

Step 1: Start the program.
Step 2: Define the required goal.
Step 3: Break the goal into smaller subgoals.
Step 4: Compare each subgoal with known facts.
Step 4: If a subgoal is satisfied, continue with the next one.
Step 5: Repeat until all subgoals are proven.
Step 6: Confirm that the original goal is true.
Step 7: Display the reasoning result.
Step 8: Stop the program.

Source Code:

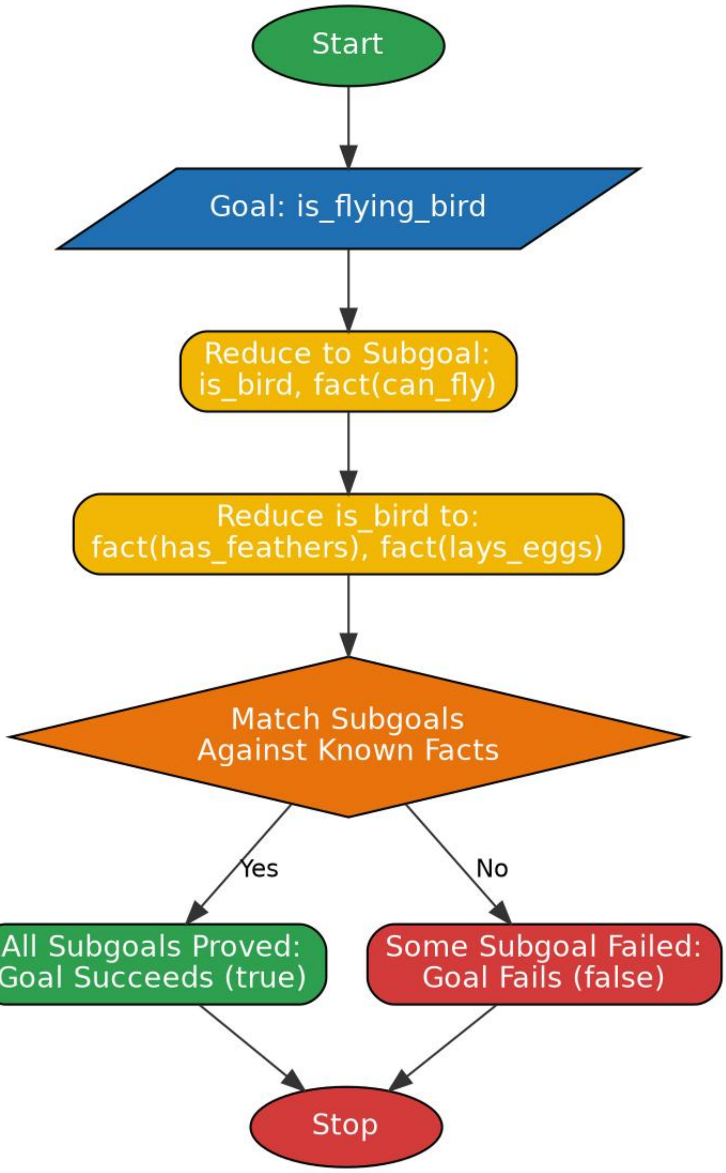
```
% Program for Backward Chaining

fact(has_feathers).
fact(lays_eggs).
fact(can_fly).

is_bird :-
   format("Checking if it is a bird...~n"),
   fact(has_feathers),
   fact(lays_eggs),
   format("Confirmed: it is a bird~n").

is_flying_bird :-
   format("Checking if it is a flying bird...~n"),
   is_bird,
   fact(can_fly),
   format("Confirmed: it is a flying bird~n").
```

Flowchart:



Start
Goal: is_flying_bird
Reduce to Subgoal:
is_bird, fact(can_fly)
Reduce is_bird to:
fact(has_feathers), fact(lays_eggs)
Match Subgoals
Against Known Facts
Yes
No
All Subgoals Proved:
Goal Succeeds (true)
Some Subgoal Failed:
Goal Fails (false)
Stop

Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [backward].
true.

?- is_bird.
Checking if it is a bird...
Confirmed: it is a bird
true.

?- is_flying_bird.
Checking if it is a flying bird...
Checking if it is a bird...
Confirmed: it is a bird
Confirmed: it is a flying bird
true.
```

Result:

Thus the Prolog program for backward chaining was written, executed successfully and the output was verified.